Calculating CPTG Galaxy Rotation Curves: DDO-154, NGC-3198, and NGC-7814

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Abstract

This tutorial presents a step-by-step calculation of the Curvature Polarization Transport Gravity galaxy-limit equation [5] using three benchmark rotation-curve systems: DDO-154, NGC-3198, and NGC-7814. Starting from SPARC-style rotation-curve inputs, the calculation converts the observed radius and velocity columns into physical acceleration fields, builds the CPTG baryonic source field, infers the galaxy-specific curvature-polarization acceleration scale, derives the structural transport scale, computes the transport remnant, solves the implicit nonlinear acceleration equation, and recovers the model rotation curve.

The three galaxies are treated separately. DDO-154 is a gas-rich dwarf system and a classic low-acceleration test case for dark matter and modified-gravity theories. NGC-3198 is a large spiral galaxy with an extended, nearly flat H I rotation curve and has long served as a standard reference case for disk-galaxy mass discrepancies. NGC-7814 is an edge-on, bulge-dominated spiral whose rotation-curve decomposition includes a large stellar bulge contribution. Together, the three examples test whether the same CPTG galaxy equation can operate across a low-mass dwarf regime, an extended disk regime, and a centrally concentrated bulge regime without changing the underlying calculation procedure.

1 Core Galaxy-Limit Equation

The reduced CPTG [5] galaxy equation is an implicit nonlinear relation for the total radial acceleration field:

$$g(r) = g_{\mathrm{bar,CPTG}}(r) + \frac{\sqrt{a_\star g_{\mathrm{bar,CPTG}}(r)} + g_{\mathrm{geom}}(r)}{\left(1 + (g(r)/a_\star)^{123/50}\right)^{11/50}}. \tag{1}$$

The geometric transport remnant is

$$g_{\mathrm{geom}}(r) = \eta a_\star \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}. \tag{2}$$

The galaxy-limit transport coefficient is fixed as

$$\eta = \frac{1}{2}. \tag{3}$$

The structural transport scale is derived from the inferred polarization scale:

$$R_c = \frac{48\pi c^2}{a_\star}. \tag{4}$$

In this tutorial, the polarization scale is treated as a galaxy-specific curvature-polarization scale inferred from the rotation-curve solution. The transport scale is then derived directly from that inferred value rather than fitted independently.

2 Input Data and Unit Conversion

Each rotation-curve file is read as a SPARC-style table following the database conventions of Lelli, McGaugh, and Schombert [6]. The required columns are radius, observed velocity, velocity uncertainty, gas velocity contribution, stellar disk velocity contribution, and stellar bulge velocity contribution. In symbolic form, the required columns are

$$r_{\rm kpc}, \qquad V_{\rm obs}, \qquad \sigma_V, \qquad V_{\rm gas}, \qquad V_{\rm disk}, \qquad V_{\rm bulge}. \tag{5}$$

The radius is converted to SI units by

$$r = r_{\rm kpc} \left(3.085677581491367 \times 10^{19}\right) \ \text{m}. \tag{6}$$

Velocities are converted by

$$V = V_{\rm km/s} \left(10^3\right) \ \text{m s}^{-1}. \tag{7}$$

The observed centripetal acceleration is then

$$g_{\rm obs}(r) = \frac{V_{\rm obs}^2(r)}{r}. \tag{8}$$

This observed field is not used as an input to solve CPTG. It is used only for comparison after the theoretical field has been calculated.

3 Constructing the CPTG Baryonic Source

CPTG does not use the same baryonic source mapping as the conventional MOND comparison field. The CPTG galaxy-limit source is built from the component velocity fields using fixed geometric weights. When a SPARC component velocity carries a sign, the signed contribution is preserved through the signed-square convention

$$Q(V) = V|V|. \tag{9}$$

For positive component velocities this reduces to the ordinary squared term:

$$Q(V) = V^2. \tag{10}$$

The CPTG source field is then constructed as

$$V_{\mathrm{bar,CPTG}}^2(r) = \frac{27}{50}Q\left(V_{\mathrm{gas}}(r)\right) + \frac{23}{50}Q\left(V_{\mathrm{disk}}(r)\right) + \frac{16}{25}Q\left(V_{\mathrm{bulge}}(r)\right). \tag{11}$$

The CPTG baryonic acceleration field is therefore

$$g_{\mathrm{bar,CPTG}}(r) = \frac{V_{\mathrm{bar,CPTG}}^2(r)}{r}. \tag{12}$$

For comparison only, the MOND-style baryonic field uses unit component weights in the standard MOND benchmark sense [7]:

$$V_{\mathrm{bar,MOND}}^2(r) = Q\left(V_{\mathrm{gas}}(r)\right) + Q\left(V_{\mathrm{disk}}(r)\right) + Q\left(V_{\mathrm{bulge}}(r)\right). \tag{13}$$

The corresponding MOND baryonic acceleration is

$$g_{\mathrm{bar,MOND}}(r) = \frac{V_{\mathrm{bar,MOND}}^2(r)}{r}. \tag{14}$$

The separation between the two source fields matters. In CPTG, the component velocities are used once to build the CPTG baryonic source. After that, the nonlinear CPTG equation is solved for the total field directly.

4 Solving the Galaxy Equation

4.1 Step 1: Determine the Polarization Scale

For a trial value of the curvature-polarization acceleration scale, the structural transport scale is first computed:

$$R_c = \frac{48\pi c^2}{a_\star}. \tag{15}$$

The nonlinear galaxy equation is then solved for the total acceleration field. The predicted circular velocity follows from

$$V_{\mathrm{CPTG}}(r) = \sqrt{g(r)r}. \tag{16}$$

The best value of the polarization scale is the value that minimizes the rotation-curve residual statistic:

$$\chi^2_{\mathrm{CPTG}} = \sum_i \left(\frac{V_{\mathrm{obs},i} - V_{\mathrm{CPTG},i}}{\sigma_{V,i}} \right)^2. \tag{17}$$

4.2 Step 2: Solve the Implicit Field Equation

At each radius, the acceleration field appears on both sides of the equation. It is therefore solved iteratively. In the tutorial sections below, the baryonic term is kept explicitly as the CPTG source field rather than shortened to a generic symbol.

A useful way to write the iteration is

$$g_{n+1}(r) = g_{\mathrm{bar,CPTG}}(r) + \left(\sqrt{a_\star g_{\mathrm{bar,CPTG}}(r)} + g_{\mathrm{geom}}(r) \right) \Sigma_n(r). \tag{18}$$

The nonlinear suppression factor at iteration step number n is

$$\Sigma_n(r) = \left(1 + \left(\frac{g_n(r)}{a_\star} \right)^{123/50} \right)^{-11/50}. \tag{19}$$

The dominant curvature-polarization enhancement is

$$g_{\text{pol}}(r) = \sqrt{a_\star g_{\text{bar,CPTG}}(r)}. \tag{20}$$

The total nonlinear enhancement added to the CPTG baryonic source is

$$\Delta g(r) = (g_{\text{pol}}(r) + g_{\text{geom}}(r))\,\Sigma(r). \tag{21}$$

The final CPTG acceleration field is

$$g_{\text{CPTG}}(r) = g_{\text{bar,CPTG}}(r) + \Delta g(r). \tag{22}$$

5 Reading the Structural Mode After the Solve

After the CPTG field has been solved, one may calculate the reduced curvature-structure invariant:

$$C(r) = g^2(r) + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr}\right)^2. \tag{23}$$

The outer resolved galaxy size is

$$R = \max(r). \tag{24}$$

The curvature-structure weight is

$$w(r) = \left|\frac{dC}{dr}\right|. \tag{25}$$

The characteristic structural length is

$$\lambda = \frac{\int_{R/5}^{R} r w(r)\,dr}{\int_{R/5}^{R} w(r)\,dr}. \tag{26}$$

The structural mode is

$$N = \frac{R}{\lambda}. \tag{27}$$

This diagnostic is computed after the CPTG solution. In the benchmark workflow, the solved structural-mode value is reported as the Curvature Structure Mode

Index, or CSMI. The numerical CSMI value can then be mapped into a Mode-type label, so the index provides both a continuous structural measure and a discrete mode-family interpretation. It is not part of the MOND comparison and is not used to construct the MOND field.

6 Example I: DDO-154 as a Low-Acceleration Dwarf

6.1 Input Profile and Solved Scale

DDO-154 is a low-mass, gas-rich dwarf galaxy whose rotation curve has long been used as a sharp test of dark matter and modified-gravity models [3, 8]. Its stellar contribution is small, its gas contribution is dynamically important, and it sits deep in the low-acceleration regime. The SPARC mass-model entry for DDO-154 [6] gives a distance of 4.04 Mpc, with 12 rotation-curve points spanning 0.49 kpc to 5.92 kpc. The bulge component is zero throughout this SPARC rotation-curve decomposition:

$$V_{\text{bulge}}(r) = 0. \tag{28}$$

The CPTG source field is therefore built from gas and disk only:

$$V^2_{\text{bar,CPTG}}(r) = \frac{27}{50} V^2_{\text{gas}}(r) + \frac{23}{50} V^2_{\text{disk}}(r). \tag{29}$$

The best-fit CPTG curvature-polarization scale is

$$a_\star = 1.497936844 \times 10^{-10} \text{ m s}^{-2}. \tag{30}$$

The derived structural transport scale is

$$R_c = 9.047717100 \times 10^{28} \text{ m}. \tag{31}$$

6.2 Representative Calculation Rows

Representative radii from the DDO-154 calculation are shown in Table 1. All acceleration fields in the table are in meters per second squared.

Table 1: Representative DDO-154 CPTG calculation rows.

Radius (kpc)	V_{obs}	V_{CPTG}	err	V_{gas}	V_{disk}	V_{bulge}	g_{obs}	$g_{\text{bar,CPTG}}$	g_{geom}	Σ	g_{pol}	Δg
0.49	13.80	22.21	1.60	3.74	12.31	0.00	1.259539e-11	5.109843e-12	1.537001e-24	9.948927e-01	2.766627e-11	2.752497e-11
3.46	44.60	44.54	0.20	17.60	8.37	0.00	1.863130e-11	1.868568e-12	2.371954e-23	9.987092e-01	1.673020e-11	1.670861e-11
5.92	45.50	46.65	1.30	15.64	6.26	0.00	1.133315e-11	8.217756e-13	5.030950e-23	9.995664e-01	1.109490e-11	1.109009e-11

At the outer listed radius, the CPTG baryonic source field is

$$g_{\mathrm{bar,CPTG}} = 8.217756 \times 10^{-13}\ \mathrm{m\ s^{-2}}. \tag{32}$$

The solved field is

$$g_{\mathrm{CPTG}} = 1.191186 \times 10^{-11}\ \mathrm{m\ s^{-2}}. \tag{33}$$

The missing response is supplied primarily by the nonlinear polarization term.

6.3 Observed-Versus-Calculated Velocity Check

At the outer listed radius of

$$r = 5.92\ \mathrm{kpc}, \tag{34}$$

the observed velocity is

$$V_{\mathrm{obs}} = 45.50\ \mathrm{km\ s^{-1}}, \tag{35}$$

and the calculated CPTG velocity is

$$V_{\mathrm{CPTG}} = 46.65\ \mathrm{km\ s^{-1}}. \tag{36}$$

6.4 Fit Summary and Interpretation

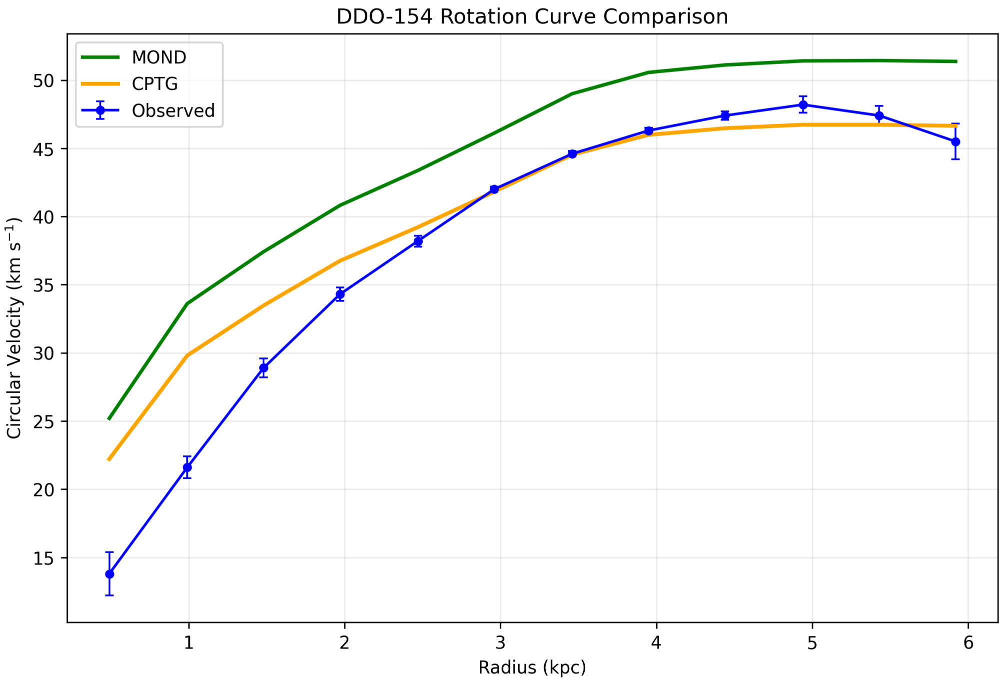

Figure 1: Rotation-curve comparison for DDO-154. The blue curve shows the observed data, the green curve shows the MOND comparison, and the orange curve shows the CPTG fit.

Table 2: DDO-154 fit summary.

Quantity	Value
Points	12
CPTG chi2	226.615133
CPTG chi2 per point	18.884594
MOND chi2	2359.131043
MOND chi2 per point	196.594254
CPTG RMS residual	3.774060 km s-1
MOND RMS residual	6.740476 km s-1
CPTG mean absolute error	2.453007 km s-1
MOND mean absolute error	6.101620 km s-1
Structural mode N	1.890702489
CSMI type	LSB Dwarf Disk

DDO-154 is a strong low-acceleration test because its baryonic field is weak. CPTG handles this through the square-root polarization response. The calculation demonstrates that the galaxy equation does not simply add an arbitrary halo-like term. Instead, it constructs the CPTG-weighted baryonic source field, infers a single system scale, and solves the implicit nonlinear field equation.

For this galaxy, the direct transport remnant is not the numerically dominant term at measured radii. The visible effect is curvature polarization. The structural mode diagnostic then reads back the curvature organization from the solved field and classifies the object as a low-mode dwarf-disk system.

7 Example II: NGC-3198 as an Extended Flat-Curve Spiral

7.1 Input Profile and Solved Scale

NGC-3198 is one of the classic extended rotation-curve galaxies, and its H I curve has long served as a standard benchmark for both dark-halo modeling and modified-gravity tests [1, 2]. The SPARC mass-model entry for NGC-3198 [6] gives a distance of 13.8 Mpc, with 43 rotation-curve points spanning 0.32 kpc to 44.08 kpc. The bulge component is zero throughout this SPARC rotation-curve decomposition:

$$V_{\mathrm{bulge}}(r) = 0. \tag{37}$$

The CPTG source field is again built from gas and disk contributions:

$$V_{\mathrm{bar,CPTG}}^2(r) = \frac{27}{50} V_{\mathrm{gas}}^2(r) + \frac{23}{50} V_{\mathrm{disk}}^2(r). \tag{38}$$

The best-fit CPTG curvature-polarization scale is

$$a_\star = 9.858387217 \times 10^{-11} \text{ m s}^{-2}. \tag{39}$$

The derived structural transport scale is

$$R_c = 1.374759228 \times 10^{29} \text{ m}. \tag{40}$$

7.2 Representative Calculation Rows

Representative radii from the NGC-3198 calculation are shown in Table 3. All acceleration fields in the table are in meters per second squared.

Table 3: Representative NGC-3198 CPTG calculation rows.

Radius (kpc)	V_{obs}	V_{CPTG}	err	V_{gas}	V_{disk}	V_{bulge}	g_{obs}	$g_{\mathrm{bar,CPTG}}$	g_{geom}	Σ	g_{pol}	Δg
0.32	24.40	51.11	35.90	0.00	63.28	0.00	6.029470e-11	1.865479e-10	3.101411e-25	5.753058e-01	1.356120e-10	7.801835e-11
7.06	146.00	146.06	1.57	12.87	140.94	0.00	9.784766e-11	4.235470e-11	2.358717e-23	8.601002e-01	6.461803e-11	5.557799e-11
44.08	149.00	151.42	3.00	47.84	61.63	0.00	1.632226e-11	2.193171e-12	3.063992e-22	9.971684e-01	1.470412e-11	1.466249e-11

At the outer listed radius, the CPTG baryonic source field is

$$g_{\mathrm{bar,CPTG}} = 2.193171 \times 10^{-12} \text{ m s}^{-2}. \tag{41}$$

The solved total field is

$$g_{\mathrm{CPTG}} = 1.685566 \times 10^{-11} \text{ m s}^{-2}. \tag{42}$$

7.3 Observed-Versus-Calculated Velocity Check

The resulting velocity is

$$V_{\mathrm{CPTG}} = 151.42 \text{ km s}^{-1}. \tag{43}$$

The corresponding observed velocity is

$$V_{\mathrm{obs}} = 149.00 \text{ km s}^{-1}. \tag{44}$$

7.4 Fit Summary and Interpretation

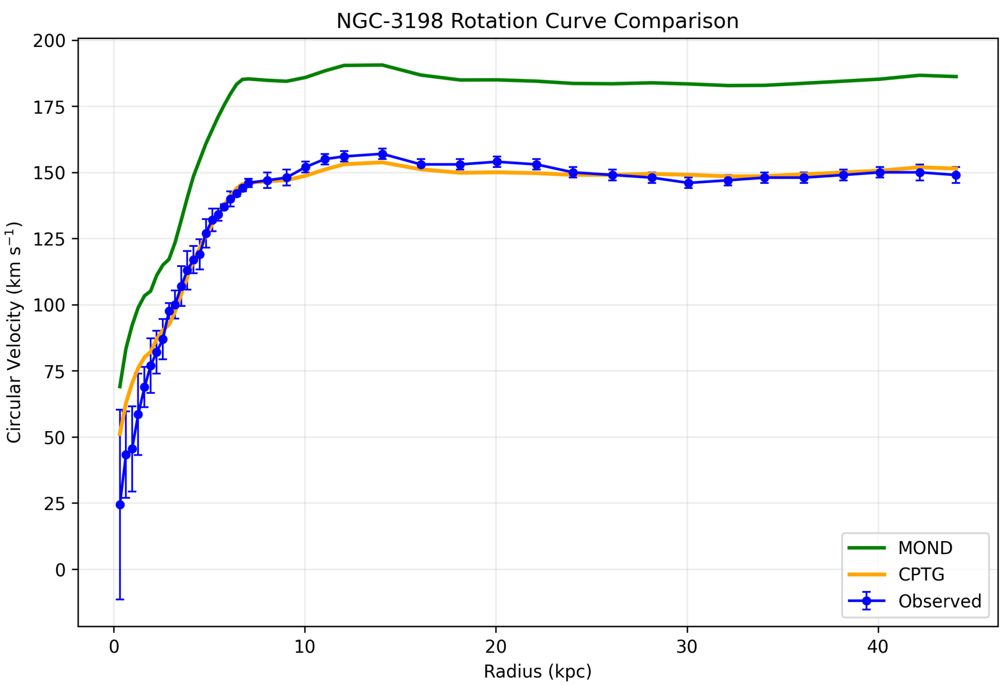

Figure 2: Rotation-curve comparison for NGC-3198. The blue curve shows the observed data, the green curve shows the MOND comparison, and the orange curve shows the CPTG fit.

Table 4: NGC-3198 fit summary.

Quantity	Value
Points	43
CPTG chi2	46.789053
CPTG chi2 per point	1.088118
MOND chi2	12186.634554
MOND chi2 per point	283.410106
CPTG RMS residual	7.449345 km s-1
MOND RMS residual	35.003291 km s-1
CPTG mean absolute error	4.171527 km s-1
MOND mean absolute error	34.617001 km s-1
Structural mode N	3.062572658
CSMI type	Early Spiral from Intermediate Spiral

NGC-3198 demonstrates the large-disk version of the same calculation. The inner disk produces a stronger baryonic source field than DDO-154, so the nonlinear suppression factor is more active in the inner galaxy. Farther out, the baryonic source weakens, the suppression relaxes, and the square-root polarization response supplies the extended acceleration needed for the nearly flat outer curve.

The measured flat regime in NGC-3198 should be distinguished from the farther asymptotic outer continuation of the CPTG solution. Current measured radii still probe the observed transition regime rather than the formal extended continuation with the asymptotic quarter-power velocity scaling.

The direct transport remnant remains small across the measured radial range because the derived transport scale is very large. In the galaxy limit, that is expected: transport appears as a coarse-grained structural remnant rather than as the displaced-curvature effect seen in merger-scale systems.

8 Example III: NGC-7814 as a Bulge-Dominated Case

8.1 Input Profile and Solved Scale

NGC-7814 is a centrally concentrated, bulge-dominated spiral whose light and mass decomposition make it a useful bulge benchmark in alternative-gravity discussions [4]. Unlike DDO-154 and NGC-3198, the SPARC mass-model entry for NGC-7814 [6] contains a large, nonzero stellar bulge velocity column across the measured radial range. The SPARC entry gives a distance of 14.4 Mpc, with 18 rotation-curve points spanning 0.63 kpc to 19.53 kpc. The bulge component is nonzero throughout this SPARC rotation-curve decomposition and dominates the inner decomposition. The first listed radius has

$$V_{\rm bulge} = 345.22 \text{ km s}^{-1}. \tag{45}$$

The outer listed radius still has

$$V_{\rm bulge} = 105.49 \text{ km s}^{-1}. \tag{46}$$

The CPTG source field therefore uses all three weighted components:

$$V^2_{\rm bar,CPTG}(r) = \frac{27}{50} Q\left(V_{\rm gas}(r)\right) + \frac{23}{50} Q\left(V_{\rm disk}(r)\right) + \frac{16}{25} Q\left(V_{\rm bulge}(r)\right). \tag{47}$$

The inner gas entries in the SPARC decomposition are signed. The signed-square convention preserves that information inside the weighted source sum, so a negative gas entry subtracts from the intermediate source budget rather than flipping the final scalar radial acceleration field by itself. In Table 5, the negative inner gas term is therefore offset by the much larger positive disk and bulge terms, leaving the net quantity $g_{\rm bar,CPTG}$ positive after division by radius.

The best-fit CPTG curvature-polarization scale is

$$a_\star = 1.830618925 \times 10^{-10} \text{ m s}^{-2}. \tag{48}$$

The derived structural transport scale is

$$R_c = 7.403457168 \times 10^{28} \text{ m}. \tag{49}$$

8.2 Representative Calculation Rows

Representative radii from the NGC-7814 calculation are shown in Table 5. All acceleration fields in the table are in meters per second squared.

Table 5: Representative NGC-7814 CPTG calculation rows.

Radius (kpc)	$V_{\rm obs}$	$V_{\rm CPTG}$	err	$V_{\rm gas}$	$V_{\rm disk}$	$V_{\rm bulge}$	$g_{\rm obs}$	$g_{\rm bar,CPTG}$	$g_{\rm geom}$	Σ	$g_{\rm pol}$	Δg
0.63	265.00	282.96	15.00	-0.69	39.69	345.22	3.612440e-09	3.960827e-09	3.536121e-24	1.854231e-01	8.515142e-10	1.578904e-10
10.65	219.00	219.96	2.90	16.09	105.99	142.79	1.459446e-10	5.585805e-11	1.852421e-22	9.036122e-01	1.011211e-10	9.137428e-11
19.53	214.00	207.20	4.87	22.64	77.35	105.49	7.599320e-11	1.684440e-11	4.329457e-22	9.796191e-01	5.552988e-11	5.439813e-11

At the first listed radius, the CPTG baryonic source field is already large:

$$g_{\rm bar,CPTG} = 3.960827 \times 10^{-9} \text{ m s}^{-2}. \tag{50}$$

The solved field is

$$g_{\mathrm{CPTG}} = 4.118717 \times 10^{-9} \text{ m s}^{-2}. \tag{51}$$

In this high-field inner region, the nonlinear suppression factor is

$$\Sigma = 0.185423. \tag{52}$$

8.3 Observed-Versus-Calculated Velocity Check

At the representative middle listed radius of

$$r = 10.65 \text{ kpc}, \tag{53}$$

the observed velocity is

$$V_{\mathrm{obs}} = 219.00 \text{ km s}^{-1}, \tag{54}$$

and the calculated CPTG velocity is

$$V_{\mathrm{CPTG}} = 219.96 \text{ km s}^{-1}. \tag{55}$$

8.4 Fit Summary and Interpretation

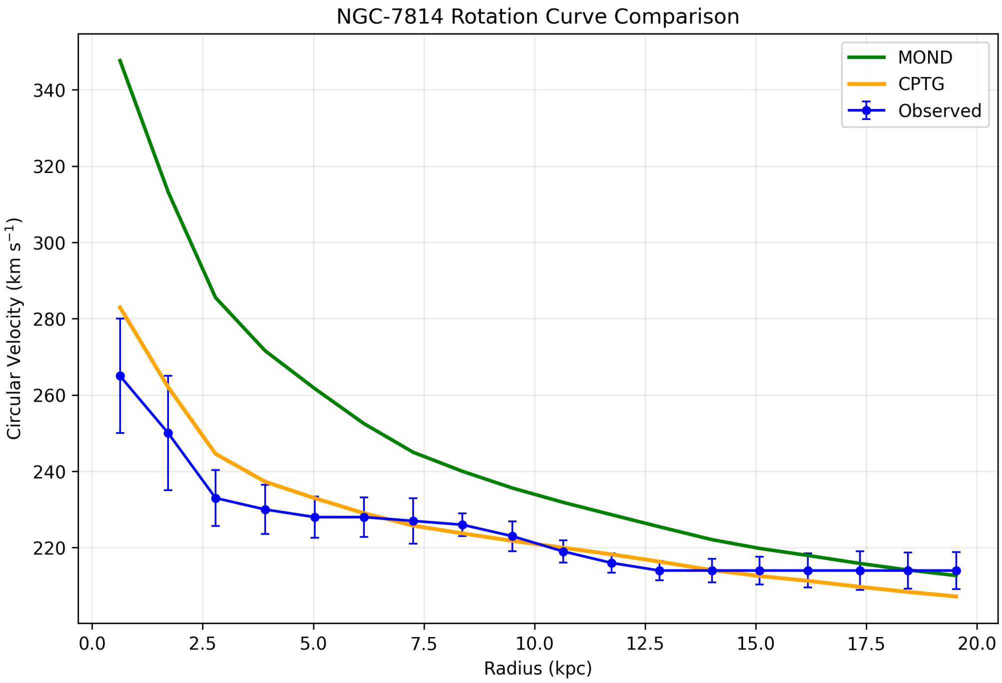

Figure 3: Rotation-curve comparison for NGC-7814. The blue curve shows the observed data, the green curve shows the MOND comparison, and the orange curve shows the CPTG fit.

Table 6: NGC-7814 fit summary.

Quantity	Value
Points	18
CPTG chi2	13.762524
CPTG chi2 per point	0.764585
MOND chi2	319.814608
MOND chi2 per point	17.767478
CPTG RMS residual	6.687395 km s-1
MOND RMS residual	31.882528 km s-1
CPTG mean absolute error	4.780270 km s-1
MOND mean absolute error	22.268552 km s-1
Structural mode N	3.157080935
CSMI type	Early Spiral

NGC-7814 demonstrates the bulge-dominated version of the same calculation. In the inner galaxy, the large bulge term produces a strong CPTG baryonic source field, and the nonlinear suppression factor prevents the polarization term from over-amplifying an already strong field. In the outer galaxy, the source field weakens and the polarization contribution becomes more important.

This makes NGC-7814 a useful complement to DDO-154 and NGC-3198. DDO-154 shows the gas-rich dwarf limit. NGC-3198 shows the extended disk limit.

NGC-7814 shows that the same equation can include a large, explicitly weighted bulge component without changing the calculation sequence.

9 Cross-Case Summary

The three-galaxy worked benchmark gives 73 total datapoints. The summary comparison is shown in Table 7. For each galaxy, one representative observed-versus-calculated velocity pair is also listed so the tutorial comparison is visible in one place.

Table 7: Cross-case tutorial summary.

Galaxy	Points	Rep. radius	V_{obs}	V_{CPTG}	$a_\star$	R_c in meters	CPTG chi2 per point	MOND chi2 per point	Structural mode
DDO-154	12	5.92	45.50	46.65	1.497936844e-10	9.047717100e28	18.884594	196.594254	1.890702489
NGC-3198	43	44.08	149.00	151.42	9.858387217e-11	1.374759228e29	1.088118	283.410106	3.062572658
NGC-7814	18	10.65	219.00	219.96	1.830618925e-10	7.403457168e28	0.764585	17.767478	3.157080935

The three examples are intentionally not redundant. DDO-154 shows the weak-source dwarf limit. NGC-3198 shows the extended disk regime. NGC-7814 shows the bulge-dominated regime. The tutorial point is that the same method is used in all three cases, while the relative importance of the source terms, the observed-versus-calculated velocity match, and the suppression factor changes from one galaxy to the next.

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